The Perfect Coffee Price: Maximizing Coffee Shop Profit Based Solely on Demand

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How to Read This Report

Sections 1 through 4 present the problem, the results, and the business recommendation. These sections are intended for readers primarily interested in the practical conclusions of the analysis.

The remaining sections provide the mathematical methodology, model derivation, and uncertainty analysis for readers interested in the technical details.

1 Context

A Coffee Shop has a table with data about price tested and demand generated by each price. The higher the price, the less the demand. The less the price, the higher the demand. The biggest question is: What is the perfect coffee price to maximize revenue?

2 Mathematical Modeling

I used a simple linear regression to model the relation between price and units sold. The line of best fit found was:

$$units = -18.30 \cdot price + 136.79$$

This line is an approximation of customer behavior. With it, I calculated the **expected revenue** for every price.

As the revenue function has the parabolic form with downward concavity, it has a single maximum point. The optimal price was identified by the peak of this curve.

3 The Solution

The model indicated to the optimal price \$3.74. However the analysis went deeper. Analyzing the revenue for some price ranges, the difference in the revenue generated by each price was minimum for prices in the range \$3.60 and \$3.80.

This is the indifference zone. The revenue curve is very flat at its peak. Adjusting the price \$0.10 higher or lower has almost no impact on the final revenue.

The following chart illustrates the revenue curve and highlights the indifference zone:

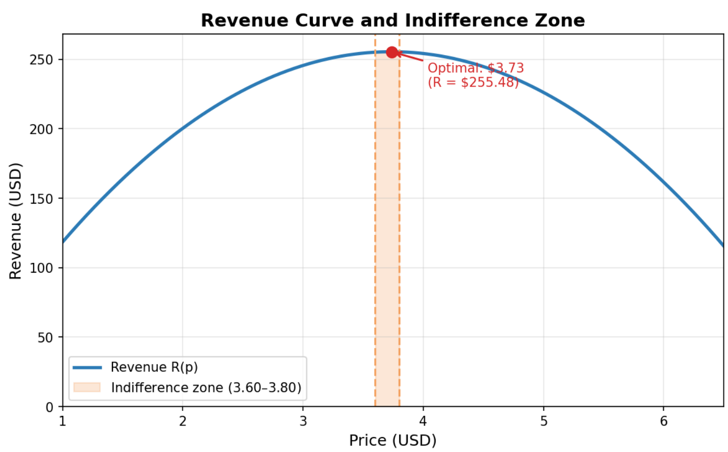

Figure 1: Revenue curve and indifference zone

4 Strategic Recommendation

Based on the model, the recommendation to the owner of the Coffee Shop is not a unique price — it's a strategic price range.

Recommendation: Keep the coffee price between \$3.60 and \$3.80. In this range, the revenue is maximized. Use this knowledge to plan promotions with margin, without being afraid of "leaving money on the table".

Note: The analysis above provides all the information needed for decision-making. The following sections cover the technical methodology for transparency and professional documentation purposes.

5 Modeling Process

The first step is to define the goal, which is to find the revenue equation described by the relation:

$$Revenue = Price \cdot Units.$$

Therefore, if the equation that models the number of units sold as a function of price is found, the revenue function can be obtained directly.

We have a simple linear regression problem, so the line of best fit will be used as an approximation of the data shown in the table.

The table below contains the observed data:

Price	Units Sold
2.0	100
3.0	85
4.0	60
5.0	45
6.0	30
2.5	95
3.5	70
4.5	50
5.5	35
6.5	20

Table 1: Observed sales data.

Let u_i denote the number of units sold and p_i the corresponding price for the i-th observation. Assuming a linear relationship between these variables, we model the demand function as

$$u_i = ap_i + b$$

where a and b are unknown parameters to be determined from the data. The model can be expressed in matrix form as

$$\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{10} \end{pmatrix} = \begin{pmatrix} p_1 & 1 \\ p_2 & 1 \\ \vdots & \vdots \\ p_{10} & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}$$

For convenience, define

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{10} \end{pmatrix}, \qquad A = \begin{pmatrix} p_1 & 1 \\ p_2 & 1 \\ \vdots & \vdots \\ p_{10} & 1 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} a \\ b \end{pmatrix}.$$

Then the model can be written compactly as

$$\mathbf{u} = A\mathbf{x}$$

In general, the system is overdetermined and does not admit an exact solution. Therefore, the least-squares approach is used, leading to the normal equations

$$A^T A\mathbf{x} = A^T\mathbf{u}.$$

Solving this system yields the coefficients of the regression line. Solving this system quickly with python, we had that the vector $\vec{x}$ is

```python
import numpy as np

# Data
price = np.array([2.00, 3.00, 4.00, 5.00, 6.00, 2.50, 3.50,
    4.50, 5.50, 6.50])
units = np.array([100, 85, 60, 45, 30, 95, 70, 50, 35, 20])

# The matrix A (with 1s column)
A = np.column_stack([price, np.ones_like(price)])

# Equation
p = np.linalg.solve(A.T @ A, A.T @ units)
a, b = p[0], p[1]

print(f"a = {a:.2f}, b = {b:.2f}")
```

Listing 1: Calculation of the coefficients of the regression line

On the output we have the vector $\vec{x}$ equals to

$$\begin{pmatrix} -18.30 \\ 136.79 \end{pmatrix}$$

With this vector, the units function will be

$$u_i = -18.30p_i + 136.79$$

As the first goal was to find the revenue function $revenue = prices \cdot units$, multiplying the price on the units function we obtain:

$$R(p) = -18.30p^2 + 136.79p$$

This is a quadratic function with downward concavity, which implies the existence of a unique price that maximizes revenue. Differentiating the revenue function and setting the derivative equal to zero yields the optimal price:

$$\$3.74$$

Now, calculating the error quickly with Python:

```
# Units expected by the model
units_pred = a * price + b

# (errors) of model
residuals = units - units_pred

# RMSE
rmse = np.sqrt(np.mean(residuals**2))

print(f"RMSE = {rmse:.2f}")
```

Listing 2: Calculation of RMSE

The RMSE (Root Mean Squared Error) obtained is

$$\text{RMSE} = 2.84.$$

This indicates that the model's predictions of demand deviate, on average, about 2.84 units from the actual observed values.

6 Uncertainty and Robustness of the Recommendation

A natural question arises: how does the inherent error of the regression model affect the optimal price? The model's predictions have an average deviation of 2.84 units, measured by the RMSE. This uncertainty propagates to the revenue function.

As a heuristic approximation, the uncertainty in demand can be propagated to the revenue function through

$$R(p) \approx R_{\text{pred}}(p) \pm |p| \cdot \text{RMSE}.$$

This assumes that the demand prediction error is of the order of the RMSE and approximately independent of price. At the optimal price $p^* = \$3.74$, the predicted maximum revenue is

$$R(3.74) = 3.74 \cdot (-18.30 \times 3.74 + 136.79) \approx 255.6.$$

The associated uncertainty interval is therefore

$$[255.6 - 3.74 \times 2.84,\ 255.6 + 3.74 \times 2.84] \approx [245.0,\ 266.2].$$

Now consider the revenue at the boundaries of the recommended indifference zone:

- For $p = 3.60$: predicted demand ≈ 70.9, revenue $R \approx 255.3$.
- For $p = 3.80$: predicted demand ≈ 67.3, revenue $R \approx 255.5$.

Both values lie comfortably inside the uncertainty interval of the optimum point. This means that, even accounting for the model's imprecision, any price between \$3.60 and \$3.80 produces a revenue that cannot be statistically distinguished from the maximum. The recommended price range is therefore robust: the coffee shop will operate consistently near its full revenue potential, regardless of short-term random fluctuations in demand.

7 Limitations and Future Work

While maximizing revenue provides a valuable baseline for understanding demand elasticity, it presents a critical limitation from a strategic business perspective: it assumes that the marginal cost of producing an additional unit is zero. In a real-world coffee shop scenario, each cup of coffee sold incurs variable costs, such as coffee beans, milk, packaging, and direct utilities.

To transform this analysis into a definitive financial decision tool, the next iteration of this research will transition from revenue optimization to **profit maximization**. Let c represent the marginal variable cost per unit. The profit function $L(p)$ can be mathematically modeled as:

$$L(p) = (p - c) \cdot u(p)$$

Substituting our empirical demand function $u(p) = ap + b$, the quadratic profit equation becomes:

$$L(p) = (p - c)(ap + b) = ap^2 + (b - ac)p - bc$$

Differentiating $L(p)$ with respect to price p and setting the derivative to zero yields the true profit-maximizing price:

$$L'(p) = 2ap + (b - ac) = 0 \implies p^*_{\text{profit}} = \frac{ac - b}{2a} = \frac{c}{2} - \frac{b}{2a}$$

Given our calculated parameters $a = -18.30$ and $b = 136.79$, the optimal price shifts from the revenue peak and becomes dynamically dependent on the operational cost structure c. Implementing this cost-aware optimization will prevent high-volume, low-margin traps, ensuring the business targets the point of maximum financial efficiency rather than just maximum cash flow.

8 Conclusion

A simple linear regression model was used to estimate the relationship between price and demand. The resulting revenue function was optimized analytically, leading to an estimated optimal price of \$3.74.

A sensitivity analysis based on the model's prediction error showed that prices between \$3.60 and \$3.80 generate nearly identical revenues. Therefore, the practical recommendation is to operate within this interval, where revenue remains close to its maximum while preserving pricing flexibility.